Geometric Series

For $r \neq 1$, the sum of the first n terms of a geometric series is

$$a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} = \sum_{k=0}^{n-1} ar^k = a\left(\frac{1-r^n}{1-r}\right),$$

Geometric Series

A **geometric series** is the sum of the terms in a geometric sequence. If the sequence has a definite number of terms, the simple formula for the sum is

Formula 3:

$$S_n = \frac{a_1\left(1-r^n\right)}{1-r} \quad (r \neq 1)$$

This form of the formula is used when the number of terms (n), the first term (a_1), and the common ratio (r) are known.

Another formula for the sum of a geometric sequence is

Formula 4: $S_n = \dfrac{a_1 - a_n r}{1-r} \quad (r \neq 1)$

This form requires the first term (a_1), the last term (a_n), and the common ratio (r) but does not require the number of terms (n).

Arithmetic progression

$$S_n = a_1 + (a_1 + d) + (a_1 + 2d) + \cdots + (a_1 + (n-2)d) + (a_1 + (n-1)d)$$

Summary of formulas [edit]

If

a_1 is the first term of an arithmetic progression.

a_n is the nth term of an arithmetic progression.

d is the difference between terms of the arithmetic progression.

n is the number of terms in the arithmetic progression.

S_n is the sum of n terms in the arithmetic progression.

$\overline{n}$ is the mean value of arithmetic series.

then

1. $a_n = a_1 + (n-1)d,$
2. $a_n = a_m + (n-m)d.$
3. $S_n = \frac{n}{2}[2a_1 + (n-1)d].$
4. $S_n = \frac{n}{2}(a_1 + a_n).$
5. $\overline{n} = S_n/n$
6. $\overline{n} = \frac{a_1 + a_n}{2}.$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}.$$

In general:

$$\sum_{i=1}^{n} i^m = \frac{1}{m+1}\left[(n+1)^{m+1} - 1 - \sum_{i=1}^{n}\left((i+1)^{m+1} - i^{m+1} - (m+1)i^m\right)\right]$$

$$\sum_{i=1}^{n-1} i^m = \frac{1}{m+1}\sum_{k=0}^{m}\binom{m+1}{k}B_k n^{m+1-k}.$$

Geometric series:

$$\sum_{i=0}^{n} c^i = \frac{c^{n+1}-1}{c-1}, \quad c \neq 1, \quad \sum_{i=0}^{\infty} c^i = \frac{1}{1-c}, \quad \sum_{i=1}^{\infty} c^i = \frac{c}{1-c}, \quad |c| < 1,$$

$$\sum_{i=0}^{n} ic^i = \frac{nc^{n+2} - (n+1)c^{n+1} + c}{(c-1)^2}, \quad c \neq 1, \quad \sum_{i=0}^{\infty} ic^i = \frac{c}{(1-c)^2}, \quad |c| < 1.$$

Harmonic series:

$$H_n = \sum_{i=1}^{n}\frac{1}{i}, \qquad \sum_{i=1}^{n} iH_i = \frac{n(n+1)}{2}H_n - \frac{n(n-1)}{4}.$$

$$\sum_{i=1}^{n} H_i = (n+1)H_n - n, \quad \sum_{i=1}^{n}\binom{i}{m}H_i = \binom{n+1}{m+1}\left(H_{n+1} - \frac{1}{m+1}\right).$$